



Mechanical Properties of Materials

The mechanical properties of a material refer to its **response to applied forces or loads**.

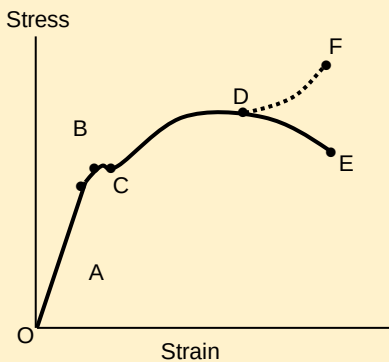
Strength

The ability of a material to withstand or resist applied forces without permanent deformation or failure.

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Stress-Strain Diagram

A diagram that shows the relationship between the stress and the strain is known as the **stress-strain diagram**.



- A → proportionality limit
- B → elastic limit
- C → yield point
- D → ultimate strength
- E → rapture strength
- F → actual rapture strength

Elastic limit : the stress beyond which the material will not return to its original shape when the load is removed

Ultimate stress/strength : the highest ordinate in the stress-strain diagram.

Yield point : the point where there is an appreciable elongation or yielding of the material even without any corresponding increase of load.

Rupture strength : the stress at failure

Resilient material : a material that absorbs and release strain energy without permanent deformation.

Hardness

The resistance of a material to indentation, scratching, or penetration. It is an important property for materials that need to resist wear, abrasion, or deformation. Hardness is commonly measured using tests such as the Rockwell hardness test or the Vickers hardness test.

Elasticity

The ability of a material to deform under applied stress and then return to its original shape when the stress is removed. The elastic modulus or Young's modulus measures a material's stiffness and its ability to resist deformation.



Ductility

The extent to which a material can undergo plastic deformation without fracturing. Ductile materials can be elongated or stretched under tensile stress without breaking. This property is important for materials that need to be shaped or formed, such as wires or sheets.

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Toughness

The combination of strength and ductility, representing a material's ability to absorb energy before fracture. It is a measure of a material's resistance to cracking or breaking under impact or sudden loads.

Fatigue Resistance

The material's ability to withstand repeated or cyclic loading without failure. It is particularly important for materials subjected to dynamic or fluctuating loads over time.

Creep Resistance

The ability of a material to resist deformation under long-term exposure to constant stress or load at elevated temperatures. It is crucial for materials used in high-temperature applications.

Fracture Toughness

A measure of a material's resistance to the propagation of cracks. It indicates how well a material can resist the growth of pre-existing flaws or cracks without sudden failure.

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Electrical Properties of Materials

Conductivity

The ability of a material to conduct electric current. Good conductors have high conductivity, allowing electrons to flow easily through them.



Resistivity

Refers to the material's resistance to the flow of electric current. Insulators have high resistivity and do not allow the flow of electric current easily.

Dielectric constant

Also known as relative permittivity, measures how well a material can store electrical energy in an electric field. It describes the ability of a material to polarize in response to an applied electric field.

Piezoelectricity

Piezoelectric materials generate an electric charge in response to applied mechanical stress, or they exhibit mechanical deformation when an electric field is applied.

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